Department of Economics

MSc Extended Essay Cover Sheet

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Intergenerational Income Mobility and Family Size: Empirical Study with OCP in Modern China

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Abstract

This research investigates the intergenerational income mobility and effect of family size on that in contemporary China. To solve the problem of edogeneity, this research uses the plausible exogenous variation in family size caused by the relaxation of the One-Child Policy (OCP) to construct instruments. Results indicate that intergenerational income elasticity in modern China is lower than that in earlier studies. Family size has a favourable effect on intergenerational income mobility, and 2SLS estimates are statistically distinct from OLS estimates. Education is a proposed channel through which family size influences the transmission of economic and non-economic traits between generations.

Keywords: Intergenerational income mobility, family-size effect, One-Child Policy

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1 Introduction

Since the 1970s, China's economy has changed from a planned to a market economy. The economy and social income have grown quickly. However, issues like economic inequality have surfaced. After the economic reform, the market took the responsibility of allocating resources. Due to the government's emphasis on labour market equality, the labour market was relatively equal before the reform. During the social transition, the government's reduced engagement in the labour market led to increased inequality.

It is believed that countries with greater economic inequality have lower intergenerational income mobility. High intergenerational income immobility is a significant factor in income inequality and undermines social justice. The transfer of wealth from parents to children is an essential component of the income dynamics of a society. Consequently, analysing intergenerational income mobility contributes significantly to our understanding of social equality.

Numerous researchers in economics and a variety of other social disciplines have long been interested in intergenerational income mobility. The framework of intergenerational investment established by Becker & Tomes (1986) has inspired hundreds of research on parental-offspring ties and the drivers of the correlation. Several characteristics, such as health (Wu and Li, 2011), parental education (Black, Devereux and Salvanes, 2005), neighbourhood (Page and Solon, 2003a), etc., have been cited in prior research as having an impact on the intergenerational transmission of income. The basic hypothesis is that these characteristics influence the development of human capital in children, hence fostering a favourable intergenerational bond between children and their parents.

However, less attention has been paid to the impact of family size on intergenerational income mobility. Family size can have a two-way effect on children's human capital formation. On the one hand, the presence of siblings, as suggested by the quantity-quality tradeoff model, can dilute parental investment—for example, siblings diminish upward mobility by decreasing an individual's years of

schooling, putting downward pressure on future income (Liu xiaoge, 2017). Conversely, the opposite may be true. For instance, raising more children can result in economies of scale (Qian, 2009). As a result, we might investigate a possible link between family size and intergenerational wealth mobility via the development of children's human capital.

Endogeneity, which mostly arises from parental variation, is one issue that has hampered the reliability of causality. For instance, if parents who place a premium on education prefer a small number of children, the tradeoff between quantity and quality will be first determined by parental preferences. In the past, to eliminate endogeneity induced by unobservable variables and simultaneity, exogenous variation in family size was frequently employed as an instrument. These instruments include indicators of birth order, twin birth (black, 2005; Angrist et al., 2006), multiple birth (Cáceres-Delpiano, 2006), and sex composition of children (Angrist et al., 2010).

This research addresses the issue of endogeneity by exploiting the regional variation in the relaxation of China's One-Child Policy (OCP). Under this policy, three factors influence the number of children a family is permitted to have: First, a person is impacted by the relaxation if he or she is born in a relaxed area; second, only girls are affected among first-born children in relaxed areas; and third, a girl is more likely to get siblings as a result of the relaxation if she has a particular household registration status. The main instrument for family size is constructed on the gender of the first birth and provincial-level fertility policy. Furthermore, families with more than the allowed number of children will result in a penalty proportional to their income. The penalty influences household decisions on fertility. This method is compelling for the reasons listed below. First, the restrictive and obligatory regulation creates a source of exogenous variation in family size, which relaxes the strong assumption of parental preferences. Secondly, the strategy makes it possible to evaluate the causal effect of an additional sibling in single-child families, when previous research has primarily focused on families with at least two children. Therefore, exploiting OCP as instruments can help in understanding the average effect of an additional child in households with various family sizes.

The results show that in the absence of family size effect, the intergenerational income elasticity in China during 2010-2018 is given as 3.1% by OLS and 2.3% by 2SLS with full controls. This research presents evidence for the negative influence of family size underlying the transmission of economic status between generations, which provides an insight consistent with earlier studies. One possible mechanism suggested is that family size may play a negative role in children's education attainment.

This research is organized as follows. Section 2 reviews the existing literature on topics including intergenerational income mobility and family size effects. Then this section discusses the potential endogeneity and earlier attempts to establish causality, creating strategy motivation and theoretical framework for research. Section 3 introduces the information concerning One-Child Policy. Section 4 presents the data source and sample construction. Section 5 establishes the identification strategy. Section 6 presents the empirical result. Section 7 provides concluding remarks.

2 Literature review

2.1 Intergenerational income mobility in China

Changes in intergenerational income mobility in China can often be separated into three phases (Gong et al., 2012). First is the traditional Chinese society (prior to 1949). In this period, the majority of social welfare was provided by the family, and there were strong ties between parents and children. Parents would invest a substantial amount of their money and assets in the education and career development of their children.

The second time period is the Maoist era (1949-1977). The early socialist reform aimed to undermine the social bonds between parents and offspring by prohibiting children from inheriting wealth from their parents and by expanding educational and occupational options for children in relatively poor families (Cheng & Dai, 1995). These factors may have facilitated intergenerational mobility. However, there are still strong familial ties, and some government policies have even reduced mobility. For example, the policies of intergenerational job replacement and internal recruitment, which were implemented in 1977 and terminated in 1986 (Yu &Liu, 2004). Moreover, the household registration system implemented under Mao's reign restricted geographic labor mobility, both from rural areas to urban regions, and from small urban cities to major urban areas, hence restricting intergenerational mobility (Wu & Treiman, 2003)

The period from the Reform and Opening-up to the present is the third period. During this time period, social mobility trends are not increasing monotonically. The change from a planned to a market-oriented economy has shifted modern China's reliance from social assistance to individual and family resources. Family networks play a significant influence in obtaining employment, which may have a negative impact on intergenerational mobility. In contrast, a steady growth in regional mobility may have increased intergenerational mobility throughout China (Wu & Treiman, 2003)

In China, intergenerational income mobility research has a significantly shorter

history than in Western nations, primarily due to insufficient data source. Few studies have been conducted on the topic.

Cheng & Dai (1995) conducted the earliest analysis of intergenerational mobility in modern China. Both men and women in China have a lower level of absolute mobility compared to their counterparts in Western industrial economies.

The results of more recent studies indicate that China's intergenerational income elasticity is relatively high, but the exact number is quite varied. Zhang & Eriksson (2010) used CHNS data from 1989 to 2006 to estimate family income over child income elasticity as 0.45. According to Gong et al. (2012), the elasticity of Chinese male's intergenerational income is 0.63, and the elasticity of female intergenerational income 0.97, indicating low intergenerational mobility in Chinese society. Deng et al. (2012) found that the intergenerational income elasticity of urban residents in 1995 and 2002 was 0.47 and 0.53, substantially higher than in other developed countries.

The difference in empirical results was derived from different datasets and regression methods. The research of Qin et al. (2014) has shown that traditional estimation methods ignore the direct transmission of human capital, which lead to underestimation. When the direct transmission of human capital was taken into account, the elasticity of China's intergenerational income has been increased from 0.429 to 0.481.

Research results about the trend of intergenerational mobility in China are also mixed. Yin (2006) observed that Chinese urbanites' economic mobility declined between 1998 and 2002 compared to 1991 to 1995. Chen &Yuan (2012) found that intergenerational income elasticity from 1988 to 2005 first declined sharply and then gradually stabilized. He & Huang (2013) suggested that intergenerational income elasticity of China showed a general trend of decline between 2000 and 2009, but family factors still play a big role in children's income. Li & Zhou (2014) found that intergenerational transmission of human capital strengthens the favourable impact of father's income on children's income. Literature shows that in the early stage of Reform and Opening up, social mobility in China has experienced a short increase but empirical evidence on the intergenerational mobility trend in China in more recent

years has been mixed.

2.2 Family-size effects

There is a large body of research on the correlation between family size and child outcomes. The quantity-quality trade-off model, introduced by Becker (1960) and developed by Becker & Lewis (1973) and Becker & Tomes (1976), provides a theoretical framework for this issue. According to the model, an increase in child quantity raises the shadow price of child quality with a given budget constraint, implying an inverse relationship between quantity and quality. Two significant implications can be drawn from the model. First, family size is an input in the child quality production function. Second, exogenous family size reductions should increase parental investment in children, improving outcomes through human capital accumulation. However, one could imagine a scenario in which siblings contribute to the formation of a child's human capital by, for example, sharing books and knowledge or decreasing the likelihood that both parents work outside the home. As a result, the family-size effect on child outcomes remains an empirical question. When data sets from different countries and different ways of identifying people are used, the results of empirical studies on the family-size effect are mixed, both in terms of signs and significance.

While there have been numerous empirical studies of the relationship between family size and the outcomes of children, there has been little prior literature that focuses on the transmission between the two generations or investigates the intergenerational income elasticity. Lindahl (2008) examined the effects of birth order and family size on intergenerational income mobility using a sample of 74,000 individuals born in Sweden between 1962 and 1964. He identified considerable birth-order and family-size patterns in the transmission of economic status from fathers to sons. In particular, the intergenerational income elasticity falls with family size and birth order within a given family size, and these discrepancies cannot be explained by changes in the ages of fathers at the time of data collection.

Feng (2018) used CFPS data from 2010 to 2014 to examine the impact of

China's family planning policy on intergenerational income mobility and found that the number of children had a positive but moderate impact. Based on CFPS2010 data, liu (2016) found that the number of children increases parental income's influence on offspring income, indicating that family planning policy may improve intergenerational mobility. Liu & Wei (2016) discovered that family planning enhanced children's education but diminished social capital as measured by sibling number. They found that family planning doesn't affect intergenerational income mobility. Kang & Wen (2019) found that parents' socioeconomic status positively affects children's socioeconomic status, and being an only child amplifies this effect.

Obviously, no consensus is achieved on whether the number of children increases intergenerational mobility. No previous studies examined how child number influences intergenerational mobility.

2.3 Endogeneity of family size

Determining the causal influence of family size on child outcomes is the primary obstacle. One issue is omitted variables. Unobservable parental characteristics, such as ability, income expectation, and sociocultural factors, can influence parents' fertility decisions and investment in children, hence affecting the outcomes of children. For example, parents with relatively lower endowment of ability and wealth might have limited knowledge and access to contraceptive methods resulting in a larger family size and a lower level of investment in their children. Highly educated parents may tend to improve the quality of their children and therefore, reduce the size of their families and concentrate on investing in one child. Ordinary least squares (OLS) estimates will be downward biased in this case. Another issue to consider is simultaneity. While family size has been shown to influence children's outcomes, the quality of previous children may also influence the desired number of children by parents. Both omitted variables and simultaneity lead to endogeneity, in which case the OLS estimate becomes biased and inconsistent.

Numerous studies attempted to identify the causal mechanism linking family size and child outcomes and to solve the endogeneity problem. A growing body of

empirical literature has adopted the instrumental variable (IV) approach, which requires a source of variation in fertility that is orthogonal to any unobservable household factors. Rosenzweig & Wolpin (1980) are the first to propose employing twin births as a tool. Due to its randomness and complete compliance, unanticipated twin births cause an external shock to a household's size. Recent estimates of twin births include those by Black et al. (2005), Cáceres-Delpiano (2006), and Angrist et al. (2006). Sex composition is an additional literary instrument created in recent literature (e.g. Angrist and Evans, 1996; Conley and Glauber, 2006; Angrist et al., 2010; Black et al., 2010; De Haan, 2010). The central premise behind these instruments is that parents with children of the same gender are more likely to have an additional child than parents with children of different genders. If children's sex composition is randomly assigned by nature, then families with several children of the same gender are more likely to have additional children.

Both of the aforementioned sources of variation have the restriction that they do not account for the influence on the marginal child. Due to the fact that the study group focuses on older children with a younger sibling, these methodologies cannot determine the effect of being born into a larger household on the younger child. Another limitation is that the implicit requirement of these instruments limits samples to families with at least two children, so the effect of increasing the number of children from one to two remains unknown.

Recently, researchers investigated China's One-Child Policy in order to revisit the quality-quantity trade-off theory. Qian (2009), for example, estimates the effect of family size on first-child school enrollment using this policy. Several policy enforcement features, such as the eligibility for having more than one child and the amount of fines for above-quota births, are exploited for the construction of instruments in this branch of research.

3 Background and One-Child Policy

In the 1970s, following two decades of explicit encouragement of population growth, Chinese policymakers adopted a series of steps to reduce population growth, particularly among the Han ethnic group. The OCP was initially formed in 1979. Its enforcement was tightened gradually until it was effectively established across the nation in 1980.

Local governments began giving permits for the birth of a second child since 1982. This practice has two main purposes. One is to stop female infanticide and sex-selection abortion. Another is to transfer responsibility from the central governments to the local and provincial government so that local governments have more power to develop policies to suit regional conditions.

According to provincial fertility policies, 31 administrative regions at the provincial level can be categorized as follows: firstly, for households with an urban household registration status, the eligible number of children is one. Second, for households with rural household registration status, there are three types of provincial fertility policies: 1. “single-child” region: all couples in six provinces are allowed to have one child: Beijing, Shanghai, Tianjin, Jiangsu, Sichuan, Chongqing; 2. "1.5 children" region: couples with rural household registration can have a second child after a four year interval if the first child is a girl in 19 provinces; 3. “two-children” region: all couples with rural household registration status are allowed to have two children in five provinces: Xinjiang, Yunnan, Ningxia, Hainan, Qinghai. Therefore, the number of children a family is eligible to have is jointly determined by the gender of the first child, household registration status, and provincial-level fertility policy.

It should be noted that OCP doesn’t mean that every household can have only one child. In general, urban birth limitations tend to be more strict than suburban birth limitations, and birth limits in eastern areas tend to be more strict than birth limits in western areas. Moreover, ethnic minorities enjoyed greater relaxation than the Han.

4 Data

4.1 Data source and sample construction

The main source of microdata for this study is the China Family Panel Study (CFPS) from 2010 to 2018. CFPS, which has been administered by Peking University's Institute of Social Science Survey (ISSS) since 2010, is a nationally representative longitudinal survey of communities, families, and individuals in China. CFPS focuses on both economic and non-economic well-being of Chinese residents and is widely used in studies of economic activity, educational outcomes, family relationships, and health in order to reflect changes in China's society, economy, population, education, and health.

The dataset from CFPS is considered as one of the best datasets available for studying intergenerational income mobility for the following reasons. First, samples in CFPS are nationally representative. CFPS began the baseline interview in 2010 with a sample of 14,960 families and 42,590 individuals from 25 provinces/cities/autonomous regions, and has conducted two-year follow-up surveys. Second, the panel structure facilitates calculating lifetime income. Individuals included in the two surveys are tracked across waves. Solid technical support and self-correcting mechanisms employed by the CFPS ensure the reliability of the income information. Finally, the survey collects a comprehensive set of demographic and socioeconomic information of all household members and their parents, children, and siblings, which enables me to generate a nationally representative sample for this research.

A CFPS survey consists of five sub-surveys: family confirmation, child, adult, common part and family economic situation. Each participant in the survey is assigned a unique ID. Each family unit was given a family identification code, and the code does not change from year to year. The family confirmation surveys allow me to track baseline samples for each round of surveys. Furthermore, CFPS provides detailed birth information (e.g. birthplace), allowing the linking of the enforcement of the OCP to every family.

Observations were derived from the 2010 CFPS family confirmation baseline data in this research. The sample set is filtered by the following conditions.

(1) The child sample consists of fully-employed individuals in child generation who are older than 18 in 2018.

(2) The parent sample consists of individuals in parental generation who are older than 18 and younger than 64 in 2018[1].

(3) The family sample with the second oldest child born before 1979 was excluded, for the reason that in these families, parents' fertility decisions should not be affected by the One-Child Policy. Twins and multiple births were also excluded from the sample set.

(4) The sample with the age gap between parents and children fewer than 20 years was excluded, because in these families, mothers do not meet the legal age for marriage.

It should be noted that the siblings of an individual can be younger than 18 in 2018. This should not be a threat because before 2015, when the One-Child Policy was further relaxed, parents in sample should complete their fertility plans. I obtained a total of 5212 father-child links from the data after processing other abnormal samples and missing values.

4.2 Variable

This section gives information of main variables used in regressions.

Firstly, annual income in CFPS is comprised of wage income, operating income, property income, transfer income, and other income. A single-year measure of income can lead to measurement error and biased estimation. The direction of the bias depends on the stage of a career. Average earnings for 4-5 years are suggested to alleviate the measurement error (Solon, 1992). To obtain the best approximation for the life-time income, in this research, the income is measured as one's average income

[1] Traditional retiring age of male employee is 65, people are less likely to remain in employment after 65 years old.

for the year 2010, 2012, 2014, 2016 and 2018 for both generations. Then the average income is inflation-adjusted to the price[1] in 2018.

Secondly, the information of family size is obtained from CFPS data and measured as the number of an individual's siblings plus one (including deceased siblings). The birth order is constructed by ordering the birth dates of children in each family.

Personal characteristics, including gender, education attainment, household registration status and ethnicity are extracted from the survey launched in 2018. Corresponding to the income measure, I use the average age from 2010 to 2018 as an age measure.

Finally, this paper exploits the relaxation of the One-Child Policy as instruments. Detailed information is extracted from the China Statistical Yearbook, including the number of children eligible for a family and the penalty for violating the fertility plan according to provincial-level policies. The measurement of the relaxation of One-Child Policy takes the value of 1/1.5/2. Penalty is measured by cash punishment in proportion to parental income.

4.3 Summary statistics

Table 1 presents summary statistics of the full sample and sub-samples grouped by family size. Beginning with the full sample, the child generation on average earns 46.2% more than the father generation, with bigger standard deviation. The child generation on average is more educated compared to their parents. Male individuals account for 62.8% in the child generation, indicating that there are more male individuals earning positive incomes than female individuals in Chinese labor force. 75.1% child samples have household registration in rural areas. Han individuals account for 92.1% in the full sample. The average age of the child generation is about 28. The average age of the father generation and mother generation are about 54 and

[1] CPI is obtained from National Bureau of Statistics of china as 101.9% in December 2018.

53 respectively. The age gap between two generations is about 26 years. On average, each father has 2.042 children in the full sample.

Then turn to the results of two generations with different family size. There are 5212 observations in total and the same number of household. 2-children household comprises 47.7% of the full sample, then followed by one-child household accounting for 28.3%, and 3-and-more-children household accounts for 23.9% of the full sample. The statistical summary reveals a possible correlation between family size and income. In the range of 42.45% to 49.60%, the child generation earns more than their father generation across all categories. The greatest dramatic increase in income between generations occurs in two-child families. In addition, in both generations, the means and variances of income drop with family size.

With reference to educational attainment, there is a negative association between educational attainment and family size. It is consistent with the identification issue that parents value a higher level of education may desire a smaller family size and place greater importance on the investment in their children.

Finally, it is noteworthy that there are fewer female samples in one-child families, while the gender ratios are more even among the full sample and other sub-samples. The imbalance might imply sex selection in some one-child families.

5 Identification strategy

This section describes the identification strategy used to evaluate the effect of family size on intergenerational income mobility. To address the issue of endogeneity, the instrumental variable approach is used. The instrument is constructed on plausible exogenous variation in family size, which is the outcome of regional heterogeneity in One-Child Policy relaxation.

Firstly, a baseline model (1) comes from Becker & Tomes (1979).

$$y_i^c = \beta_1 y_i^f + \epsilon_i \quad (1)$$

Where y_i represents an individual i's income. The superscript c means child generation, and f means father generation.

The estimation of the baseline model meets problems: first, unobservable permanent income; second, life-cycle bias. In empirical analysis, we can only obtain short-term income. The variance of short-term income is greater than that of permanent income, which leads to bias in estimation of intergenerational income elasticity. According to life cycle theory, short-term income presents an inverted U-trend with age. The error in estimating the intergenerational income elasticity caused by cyclical income changes is called life-cycle bias. In order to estimate intergenerational income elasticity with more accuracy, in model (2), I add controls and use the multi-year average income as a proxy for permanent income (Solon, 1992).

$$\ln(y_{ij}^c) = \beta_o + \beta_1 \ln(y_{ij}^f) + \gamma A_{ij} + kX_{ij} + \theta_j + \epsilon_{ij} \quad (2)$$

where $\ln(y_{ij}^c)$, the natural logarithm of an individual i's average income is a function of $\ln(y_{ij}^f)$, the natural logarithm of father's average income; A_{ij}, a vector of

age controls, including the average age and the its square; X_{ij}, a vector of individual characteristics, including children's educational attainment, father's education attainment, gender, household registration status and ethnicity; θ_j, the province fixed effects. β_1 is the intergenerational income elasticity, which measures the percentage change in offspring income relative to a marginal percentage change in parental income. $1-\beta_1$ is the intergenerational income mobility, whose value ranges between 0 and 1. If β_1 equals to 0, it means that parents' income has no influence on children's income so that the economic status in the society is fully mobile.

To examine impact of family size on intergenerational income mobility, I construct the model (3A) and model (3B) by adding family size and its interaction with father's income.

$$\ln(y^c_{ij}) = \beta_0 + \beta_1 \ln(y^f_{ij}) + \beta_2 famsize_{ij} + \gamma A_{ij} + kX_{ij} + \theta_j + \epsilon_{ij} \quad (3A)$$

$$\ln(y^c_{ij}) = \beta_0 + \beta_1 \ln(y^f_{ij}) + \beta_2 famsize_{ij} + \beta_3 \ln(y^p_{ij}) \times famsize_{ij} + \gamma A_{ij} + kX_{ij} + \theta_j + \epsilon_{ij} \quad (3B)$$

In the specification (3A), family-size effect was solely included in regression, to estimate its direct impact on income of the child generation. In the specification (3B), the interaction term between family size and father's income was included to estimate the influence of family size on the intergenerational income elasticity. the intergenerational income mobility is $1-(\beta_2+\beta_3 \times famsize)$ in (3B). If β_3 is negative, it implies that family size increases intergenerational mobility, and vice versa.

So far, the model still faces the problem of endogeneity, because the fertility decision can be endogenous. One-Child Policy provides a favorable quasi-natural experiment and causes plausibly exogenous variation in household fertility decisions. As mentioned before, family fertility decisions were exogenously affected by fertility

eligibility guided by policy and the policy varies across regions and years.

I explore two aspects of One-Child Policy to construct instruments. First, the number of children a family is eligible to have is jointly influenced by the gender of the first birth and provincial-level fertility policy. Consequently, this identification strategy is applied to the first-born children for those regions with eligibility to have a second child or more. Second, household violating the fertility plan receives penalty proportion to parental income. Therefore, instruments take the birth order, gender, regional policy and penalty into account. The first stage takes the following form:

$$famsize_{ijt} = \alpha_o + \alpha_1 ifgirl_i + \alpha_2 policy_{jt} + \alpha_3 penalty_{jt} + \alpha_4 policy_{jt} \times ifgirl_i + \alpha_5 penalty_{jt} \times ifgirl_i + \gamma A_{ij} + kX_{ij} + \theta_j + \mu_{ijt} \quad (4A)$$

$$\ln(y^f_{ij}) \times famsize_{ijt} = \alpha_o + \alpha_1 ifgirl_i + \alpha_2 policy_{jt} + \alpha_3 penalty_{jt} + \alpha_4 policy_{jt} \times ifgirl_i + \alpha_5 penalty_{jt} \times ifgirl_i + \gamma A_{ij} + kX_{ij} + \theta_j + \mu_{ijt} \quad (4B)$$

Family size of individual i, born in province j, is a function of: the $policy_{ijt}$, the extent of relaxation of One-Child Policy in province j year t, and $ifgirl_i$, a variable indicating whether the previous birth is a girl; the interaction between $policy_{ijt}$ and $ifgirl_i$; $penalty_{jt}$, the penalty of violating One-Child Policy in province j year t; the interaction between $penalty_{jt}$ and $ifgirl_i$; province-fixed effects θ_j; a vector of controls.

The validity of the IV method is contingent upon two assumptions: first, the explanatory variable is highly correlated with the instruments, and second, the instruments are exogenous to the error term. Considering the penalties for ineligible births and other types of punishment, it is plausible to assume that household fertility

decisions are substantially connected with the implementation of the policy. For the latter assumption, it can be assumed that local governments adopt provincial-level fertility strategies, which should be uncorrelated with unobservable variables at the individual level. Because the number of instrumental variables exceeds the number of endogenous variables, the 2SLS method is used in the actual regression.

6 Result

6.1 Estimates of the family size effect

The OLS and 2SLS estimations from equation (3A) are presented in Table 2. When additional control variables are included, OLS estimates for intergenerational income elasticity is about 0.032 and statistically significant. A bigger family size has a detrimental effect on child generation's income, as evidenced by all specifications. After accounting for the endogeneity of family size, the 2SLS estimates exhibit sign consistency with OLS results. In 2SLS regression with full controls, the intergenerational income elasticity decreases and statistically insignificant. The detrimental effect of family size on children's income increases in magnitude.

The OLS and 2SLS estimations from equation (3B) are presented in Table 3. When the interaction between parental income and family size is accounted for, the OLS estimates for intergenerational income elasticity increase in magnitude compared to the results from equation (3A) but is statistically insignificant. Estimates for the coefficient prior to the interaction term have negative values and are statistically significant at the 5% percent level, indicating that a bigger family size is associated with lower intergenerational elasticity and higher intergenerational economic mobility. The 2SLS regression results show consistency in sign but have little statistical significance.

6.2 First stage: family size and One-Child Policy

Tables 4A and 4B display the outcomes of first-stage regressions. One of the assumptions underlying the validity of IV estimation is that the instrument variables must be highly correlated with the endogenous explanatory variables; otherwise, instruments can be unreliable. The correlation between family size and the number of eligible children under One-Child Policy is positive and statistically significant, according to the results. This supports the hypothesis that family sizes could increase under more relaxed policies. Family size and penalty are slightly negatively related but not statistically significant. According to the findings, weak variables are unlikely

to influence the estimation of the IV.

The results of the first stage of regressions confirm earlier analysis. It is indicated that parents with varying levels of education have different preferences for family size and fertility decisions. The relationship between a father's education degree and family size is statistically significant and negative.

6.3 Robustness

To carry out robustness analysis, I examine whether the result is stable when employing an alternative measure for economic status of two generations. Table 5 presents the corresponding regression results where original lifetime income variables are substituted by income percentile rank of individuals in both generations. According to the results, the negative effect of larger family size on intergenerational income elasticity is still robust.

7 Conclusion

This research examines the effect of family size on intergenerational income mobility. It resolves the problem of endogeneity in family size by exploiting the plausibly exogeneous variation in family size due to variance in relaxation of One-Child Policy. The results show that in the absence of family size effect, the intergenerational income elasticity in China during 2010-2018 is given as 3.2% by OLS and 2.3% by 2SLS with full controls, suggesting an up-ward bias by OLS. Then after considering the family size effect on intergenerational income elasticity, this research uses the variation in family size caused by OCP to provide evidence that an additional sibling in family decreases intergenerational income elasticity, ceteris paribus.

This research shows evidence for the negative influence of family size underlying the transmission of economic status between generations, which provides an insight consistent with earlier studies. The distinction arises from the use of the latest data set and different identification strategies. The study of the causal mechanism behind intergenerational mobility is motivated by the aim of promoting social equality. Therefore, it is meaningful to understand how family size can drive the correlation between generations.

As an important means of human capital accumulation, education plays an important role in children's income. In the Mincer wage equation, income is positively correlated with educational attainment. Therefore, we can make the conjecture that education is one way in which family size affects intergenerational income transmission. In previous analyses, family size may have a two-way effect on children's educational attainment: influenced by quantity-quality trade-offs on the one hand; and receiving potential economies of scale[1] on the other hand. To test the conjecture, I use the available data to run regressions on children's educational attainment. The regression results show that a high father's income promotes children's education attainment, but that larger family size weakens this effect. One

[1] Qian (2009) finds that among one-child families, an extra child increases first-born school enrollment by around 16 percentage points.

possible mechanism for this is that limited resources are allocated to more children, resulting in fewer resources being allocated to each child. However, conclusions of how family effect influences children's education attainment cannot not be drawn with this result and more researches are needed.

There are a few challenges to the validity of outcomes. The first is selection bias. The variance in fertility eligibility may not be assigned to provinces at random. The allocation of birth eligibility is typically influenced by local economic growth and cultural factors, which might be associated with ethnic features and undermine the orthogonality between instruments and unobserved variables. Moreover, the uneven sex ratio in one-child families given in summary statistics can disprove the concept of exogenous sex of birth if sex-selection does exist.

This study does not directly address the hypothesis that bigger family sizes are associated with lower intergenerational income elasticity and greater intergenerational income mobility, hence indicating a higher level of social equality. The findings give empirical support for the proposition that having an additional child in a family with one or no children can promote intergenerational income mobility.

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Appendices

Variable Definitions

variable	Notation	Definition
Personal information		
Income(¥)	y^c, y^f	Average income in 2010, 2012, 2014, 2016 and 2018. Adjusted to 2018 price.
Family size	*famsize*	Number of children in the family. =the number of an individual's siblings + 1, including deceased siblings.
Age	*age*	The average age from 2010 to 2018
Education	*edu_c*(children) /*edu_f*(fathers)	Indicator variable =1, if illiterate/semi-illiterate; =2, if primary school; =3, if junior school; =4, if senior school; =5, if college, and university (a bachelor's degree); =6, if higher education; =0, otherwise. Illiterate/semi-illiterate refers to individuals who do not have education or not complete primary school.
Gender	*gender*	Indicator variable=1 if male; =0 if female.
Rural	*rural*	Indicator variable=1 if household registration status is rural; =0 if urban.
Ethnicity	*ethnic*	Indicator variable=1 if ethnicity is Han; =0 otherwise.
Policy		
Ifgirl	*ifgirl*	Indicator variable=1 if the family's first birth is a girl; =0 otherwise.
Policy	*policy*	The eligible number of children for a family taking the value of 1/1.5/2.
Penalty	*penalty*	A numerical variable indicating the cash punishment.

Table 1 Summary Statistics

Variable	Family size			
	1 Child	2 Children	3+ Children	All
		Children		
Income (¥)	36249.610	29512.370	26220.670	30628.682
	(29841.120)	(25349.950)	(21611.370)	(26164.239)
Age(in 2018)	28.588	27.730	27.530	27.925
	(4.846)	(4.840)	(4.710)	(4.829)
Education	4.371	3.874	3.468	3.917
	(1.360)	(1.291)	(1.343)	(1.363)
Gender	0.704	0.622	0.550	0.628
	(0.519)	(0.485)	(0.498)	(0.501)
Rural	0.572	0.797	0.868	0.751
	(0.495)	(0.402)	(0.339)	(0.433)
Ethnicity	0.945	0.921	0.891	0.921
	(0.228)	(0.269)	(0.312)	(0.270)
		Fathers		
Income (¥)	25448.190	19721.970	18087.430	20955.692
	(23160.210)	(20283.850)	(17546.530)	(20742.273)
Age(in 2018)	54.145	54.468	53.757	54.536
	(5.478)	(5.362)	(5.555)	(5.411)
Education	2.562	2.781	2.506	2.414
	(1.139)	(1.198)	(1.102)	(1.103)
Observations	1475	2487	1250	5212

Notes: Standard deviations are shown in parentheses. The sample includes 5212 children born between 1979 and 2000 and their fathers. Variables are described in Table 1.

Table 2 Regression Results: intergenerational income mobility

	Dependent variable: children's income									
	(1)		(2)		(3)		(4)		(5)	
	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
father's income	0.031**	0.001	0.030**	0.013	0.031**	0.015	0.033**	0.022	0.032**	0.023
	(0.015)	(0.020)	(0.015)	(0.021)	(0.015)	(0.020)	(0.015)	(0.020)	(0.015)	(0.020)
family size	-0.066***	-1.029***	-0.066***	-1.037***	-0.036**	-0.944***	-0.037**	-0.941***	-0.037**	-0.940***
	(0.015)	(0.112)	(0.015)	(0.113)	(0.016)	(0.133)	(0.016)	(0.156)	(0.016)	(0.156)
age	0.040	0.141***	0.039	0.146***	0.050*	0.138***	0.049*	0.130***	0.049*	0.130***
	(0.027)	(0.040)	(0.027)	(0.040)	(0.027)	(0.038)	(0.027)	(0.039)	(0.027)	(0.039)
age^2	-0.0002	-0.002***	-0.0002	-0.002***	-0.0004	-0.002***	-0.0004	-0.002***	-0.0004	-0.002***
	(0.0005)	(0.001)	(0.0005)	(0.001)	(0.0005)	(0.001)	(0.0005)	(0.001)	(0.0005)	(0.001)
edu_c	0.142***	0.017	0.141***	0.029	0.162***	0.042*	0.166***	0.067***	0.166***	0.067***
	(0.010)	(0.020)	(0.010)	(0.019)	(0.010)	(0.022)	(0.011)	(0.022)	(0.011)	(0.022)
edu_f			0.004	0.057***	0.002	0.051***	0.004	0.038**	0.004	0.038**
			(0.012)	(0.018)	(0.012)	(0.018)	(0.012)	(0.018)	(0.012)	(0.018)
gender					0.297***	0.031	0.296***	0.030	0.297***	0.030
					(0.039)	(0.033)	(0.039)	(0.042)	(0.039)	(0.043)
rural							-0.042	- 0.231***	-0.041	-0.232***
							(0.033)	(0.054)	(0.033)	(0.054)
ethnic									0.005	0.006
									(0.007)	(0.010)
Province control	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
observations	5212	5212	5212	5212	5212	5212	5212	5212	5212	5212
R-squared	0.993	0.987	0.993	0.987	0.993	0.988	0.993	0.988	0.993	0.988
Adjusted R-squared	0.993	0.987	0.993	0.987	0.993	0.988	0.993	0.988	0.993	0.988

Notes: Regression results on specification 3A.

*p<0.1; **p<0.05; ***p<0.01

Robust S.E. are shown in the parentheses.

Table 3 Regression Results: family size effect on intergenerational income mobility

	Dependent variable: children's income									
	(1)		(2)		(3)		(4)		(5)	
	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
father's income	0.041	0.357	0.044	0.338	0.053	0.354	0.050	0.605	0.050	0.602
	(0.036)	(0.634)	(0.036)	(0.592)	(0.036)	(0.654)	(0.036)	(1.299)	(0.036)	(1.288)
family size	-0.393**	1.537	-0.397**	1.435	-0.416***	1.497	-0.407**	2.615	-0.403**	2.604
	(0.159)	(2.909)	(0.160)	(2.699)	(0.160)	(2.966)	(0.160)	(5.860)	(0.159)	(5.817)
father's income× family size	-0.034**	-0.169	-0.035**	-0.159	-0.040**	-0.166	-0.039**	-0.284	-0.039**	-0.283
	(0.017)	(0.301)	(0.017)	(0.279)	(0.017)	(0.309)	(0.017)	(0.613)	(0.017)	(0.608)
age	0.041	0.029	0.041	0.048	0.052*	0.052	0.051*	0.026	0.051*	0.024
	(0.027)	(0.209)	(0.027)	(0.183)	(0.027)	(0.183)	(0.027)	(0.371)	(0.027)	(0.363)
age^2	-0.0002	-0.0001	-0.0002	-0.0004	-0.0005	-0.0004	-0.0004	-0.001	-0.0005	-0.001
	(0.0005)	(0.004)	(0.0005)	(0.003)	(0.0005)	(0.003)	(0.0005)	(0.007)	(0.0005)	(0.007)
edu_c	0.143***	0.0001	0.142***	0.030	0.163***	0.018	0.166***	0.072	0.166***	0.068
	(0.010)	(0.044)	(0.010)	(0.028)	(0.010)	(0.040)	(0.011)	(0.055)	(0.011)	(0.049)
edu_f			0.006	0.135	0.004	0.144	0.006	0.172	0.006	0.171
			(0.012)	(0.142)	(0.012)	(0.168)	(0.012)	(0.282)	(0.012)	(0.280)
gender					0.299***	0.030	0.299***	0.090	0.299***	0.088
					(0.039)	(0.072)	(0.039)	(0.186)	(0.039)	(0.181)
rural							0.034	0.720	0.034	0.717
							(0.033)	(1.018)	(0.033)	(1.007)
ethnic									0.004	0.025
									(0.007)	(0.072)
Province control	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
observations	5212	5212	5212	5212	5212	5212	5212	5212	5212	5212
R-squared	0.993	0.969	0.993	0.971	0.993	0.968	0.993	0.931	0.993	0.931
Adjusted R-squared	0.993	0.968	0.993	0.971	0.993	0.968	0.993	0.930	0.993	0.931

Notes: Regression results on specification 3B.

*p<0.1; **p<0.05; ***p<0.01

Robust S.E. are shown in the parentheses.

Table 4A Regression Results: First stage on family size

	Dependent variable: family size				
	(1)	(2)	(3)	(4)	(5)
ifgirl	0.150	0.157	0.216	0.167	0.167
	(0.157)	(0.157)	(0.156)	(0.156)	(0.155)
policy	0.740***	0.742***	0.711***	0.578***	0.569***
	(0.214)	(0.214)	(0.214)	(0.214)	(0.214)
penalty	-0.013	-0.014	-0.014	-0.016	-0.016
	(0.017)	(0.017)	(0.017)	(0.017)	(0.017)
ifgirl*policy	0.111***	0.105	0.107	0.137	0.138
	(0.106)	(0.106)	(0.107)	(0.106)	(0.106)
penalty*policy	0.007	0.007	0.007	0.005	0.004
	(0.021)	(0.021)	(0.021)	(0.021)	(0.021)
age	0.088***	0.092***	0.091***	0.085***	0.085***
	(0.027)	(0.027)	(0.027)	(0.027)	(0.027)
age^2	-0.002***	-0.002***	-0.002***	-0.001***	-0.001***
	(0.0005)	(0.0005)	(0.0005)	(0.0005)	(0.0005)
edu_c	-0.150***	-0.137***	-0.136***	-0.113***	-0.111***
	(0.010)	(0.010)	(0.010)	(0.011)	(0.011)
edu_f		-0.049***	-0.049***	-0.038***	-0.038***
		(0.011)	(0.011)	(0.011)	(0.011)
gender			0.062***	0.052***	0.051***
			(0.009)	(0.009)	(0.009)
rural				0.210***	0.210***
				(0.028)	(0.028)
ethnic					0.013**
					(0.006)
Province control	YES	YES	YES	YES	YES
observations	5212	5212	5212	5212	5212
R-squared	0.870	0.870	0.870	0.872	0.872
Adjusted R-squared	0.869	0.869	0.869	0.871	0.871
F Statistic	901.860***	882.687***	860.565***	829.443***	829.443***

Notes: The dependent variable is family size. The estimation method is ordinary least squares (OLS)

*p<0.1; **p<0.05; ***p<0.01

Robust S.E. are shown in the parentheses.

Table 4B Regression Results: First stage on family size× father's income

	Dependent variable: family size× father's income				
	(1)	(2)	(3)	(4)	(5)
ifgirl	2.307	2.340	2.857	2.468	2.470
	(1.595)	(1.595)	(1.584)	(2.590)	(1.586)
policy	6.226***	6.204***	5.928***	4.772***	4.741***
	(2.127)	(2.130)	(2.133)	(2.127)	(2.130)
penalty	-0.132	-0.140	-0.138	-0.150	-0.153
	(0.177)	(0.177)	(0.177)	(0.176)	(0.176)
ifgirl*policy	0.539**	0.510	0.5228	0.755	0.757
	(1.074)	(1.074)	(1.074)	(1.075)	(1.072)
penalty*policy	0.044	0.041	0.043	0.031	0.028
	(0.224)	(0.223)	(0.223)	(0.221)	(0.2210
age	0.870***	0.895***	0.895***	0.841***	0.836***
	(0.274)	(0.275)	(0.275)	(0.275)	(0.275)
age^2	-0.017***	-0.017***	-0.017***	-0.016***	-0.016***
	(0.005)	(0.005)	(0.005)	(0.005)	(0.005)
edu_c	-1.157***	-1.076***	-1.074***	-0.875***	-0.866***
	(0.097)	(0.101)	(0.101)	(0.105)	(0.105)
edu_f		-0.308***	-0.309***	-0.198*	-0.197*
		(0.119)	(0.119)	(0.120)	(0.120)
gender			0.548***	0.453***	0.449***
			(0.084)	(0.088)	(0.087)
rural				1.833***	1.834***
				(0.280)	(0.280)
ethnic					0.062
					(0.052)
Province control	YES	YES	YES	YES	YES
observations	5212	5212	5212	5212	5212
R-squared	0.864	0.864	0.864	0.865	0.865
Adjusted R-squared	0.863	0.863	0.863	0.864	0.864
F Statistic	792.363***	773.246***	753.843***	741.466***	723.998***

Notes: The dependent variable is family size times father's income. The estimation method is ordinary least squares (OLS)

*p<0.1; **p<0.05; ***p<0.01

Robust S.E. are shown in the parentheses.

Table 5 Regression Results: robustness check

	Dependent variable: children's income percentile rank									
	(1)		(2)		(3)		(4)		(5)	
	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
father's income rank	0.103***	1.025	0.101***	0.990	0.096***	1.130**	0.096***	1.489**	0.096***	1.445**
	(0.033)	(0.636)	(0.033)	(0.628)	(0.032)	(0.553)	(0.033)	(0.638)	(0.033)	(0.612)
family size	-0.009	-0.112	-0.009	-0.125	-0.001	-0.061	-0.001	-0.011	-0.001	-0.002
	(0.008)	(0.175)	(0.008)	(0.172)	(0.008)	(0.149)	(0.008)	(0.168)	(0.008)	(0.163)
father's income rank× family size	-0.016	-0.505	-0.016	-0.483	-0.012	-0.548**	-0.012	-0.714**	-0.012	-0.694**
	(0.015)	(0.307)	(0.015)	(0.302)	(0.015)	(0.267)	(0.015)	(0.307)	(0.015)	(0.295)
age	0.004	0.037**	0.004	0.039**	0.008	0.035**	0.008	0.028*	0.008	0.028*
	(0.009)	(0.016)	(0.009)	(0.016)	(0.009)	(0.015)	(0.009)	(0.016)	(0.009)	(0.016)
age^2	0.0001	-0.001**	0.0001	-0.001**	0.00000	-0.001**	-0.00001	-0.0004	-0.00001	-0.0004
	(0.0002)	(0.0003)	(0.0002)	(0.0003)	(0.0002)	(0.0003)	(0.0002)	(0.0003)	(0.0002)	(0.0003)
edu_c	0.056***	0.001	0.056***	0.002	0.063***	0.007	0.063***	0.019**	0.063***	0.019**
	(0.003)	(0.008)	(0.003)	(0.007)	(0.003)	(0.009)	(0.003)	(0.009)	(0.003)	(0.009)
edu_f			0.002	0.015**	0.001	0.015**	0.001	0.010	0.001	0.010
			(0.004)	(0.007)	(0.004)	(0.007)	(0.004)	(0.007)	(0.004)	(0.007)
gender					0.106***	0.011	0.106***	0.005	0.106***	0.006
					(0.013)	(0.013)	(0.013)	(0.016)	(0.013)	(0.017)
rural							-0.001	-0.134***	-0.001	-0.132***
							(0.010)	(0.028)	(0.010)	(0.028)
ethnic									0.002	0.002
									(0.002)	(0.003)
Province control	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
observations	5212	5212	5212	5212	5212	5212	5212	5212	5212	5212
R-squared	0.995	0.960	0.993	0.967	0.993	0.968	0.993	0.931	0.993	0.931
Adjusted R-squared	0.995	0.962	0.993	0.967	0.993	0.968	0.993	0.930	0.993	0.931

Notes: *p<0.1; **p<0.05; ***p<0.01

Robust S.E. are shown in the parentheses.

Table 6 Regression Results: family size effect on children's education

	Dependent variable: children's education attainment							
	(1)		(2)		(3)		(4)	
	OLS	2SLS	OLS	2SLS	OLS	2SLS	OLS	2SLS
father's income	0.392***	0.848	0.388***	3.326***	0.236***	2.100***	0.235***	1.812**
	(0.050)	(0.652)	(0.050)	(0.885)	(0.049)	(0.802)	(0.048)	(0.734)
family size	0.380*	2.817	0.382*	14.186***	0.022	8.862**	0.049	7.584**
	(0.217)	(3.052)	(0.217)	(4.156)	(0.213)	(3.741)	(0.210)	(3.428)
father's income× family size	-0.068***	-0.400	-0.072***	-1.459***	-0.026	-0.900**	-0.028	-0.768**
	(0.022)	(0.309)	(0.022)	(0.422)	(0.022)	(0.381)	(0.022)	(0.349)
age	0.474***	0.122***	0.445***	0.361***	0.441***	0.383***	0.439***	0.387***
	(0.035)	(0.044)	(0.034)	(0.058)	(0.033)	(0.047)	(0.033)	(0.045)
age^2	-0.008***	-0.002**	-0.008***	-0.006***	-0.008***	-0.007***	-0.008***	-0.007***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
edu_f	0.319***	-0.044*	0.313***	0.283***	0.235***	0.228***	0.233***	0.228***
	(0.017)	(0.023)	(0.017)	(0.034)	(0.016)	(0.025)	(0.016)	(0.024)
gender			-0.420***	-0.319***	-0.353***	-0.257***	-0.349***	-0.251***
			(0.038)	(0.042)	(0.037)	(0.047)	(0.037)	(0.046)
rural					-0.935***	-0.944***	-0.932***	-0.962***
					(0.044)	(0.103)	(0.044)	(0.099)
ethnic							0.026***	0.039***
							(0.006)	(0.010)
Province control	YES	YES	YES	YES	YES	YES	YES	YES
observations	5212	5212	5212	5212	5212	5212	5212	5212
R-squared	0.988	0.922	0.987	0.919	0.965	0.914	0.965	0.914
Adjusted R-squared	0.988	0.921	0.987	0.919	0.964	0.914	0.965	0.914

Notes: Regression results on children's education attainment.

*p<0.1; **p<0.05; ***p<0.01

Robust S.E. are shown in the parentheses.